EES 301: Global biogeochemical cycles and Climate change

Coupled reservoir systems

- **The treatment of times scales and dynamic behavior of single reservoirs can be generalized to systems of two or more reservoirs**

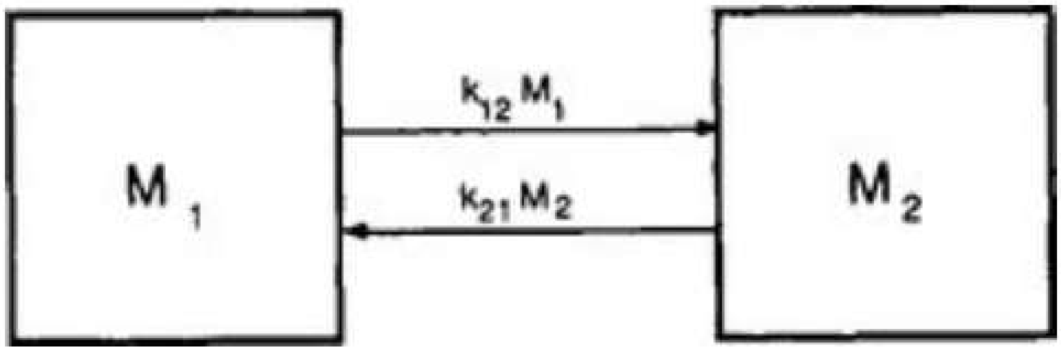


FIG. 4-6 A coupled two-reservoir system with fluxes proportional to the content of the emitting reservoirs.

- **A very important group of coupled reservoirs which are mathematically easily treatable are so called linear reservoir systems. We generally use matrices to describe such systems.**
- **When the flux is proportional to the content of the emitting reservoir the flux F_{ij} from reservoir i to reservoir j is:**

$$F_{ij} = \mathrm{k}_{\mathrm{ij}}\mathrm{M}_{\mathrm{i}}$$

Linear coupled reservoir systems

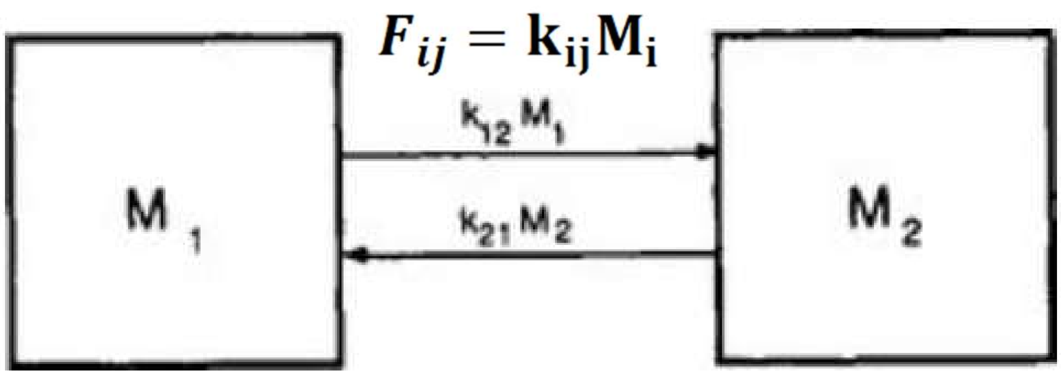


FIG. 4-6 A coupled two-reservoir system with fluxes proportional to the content of the emitting reservoirs.

- The rate of change of the amount M_i in reservoir i is thus:

$$\frac{dM_i}{dt} = \sum_{j}^{n} k_{ji}M_j - M_i \sum_{i}^{n} k_{ji} \quad for\ j \neq i$$

where n is the total number of reservoirs in the system, The vector M is equal to M_1, M_2, … M_n and the elements of matrix k are linear combinations of the coefficients k_{ij}

$$\frac{dM}{dt} = [k_{ij}]\vec{M}$$

Response time of coupled reservoir systems

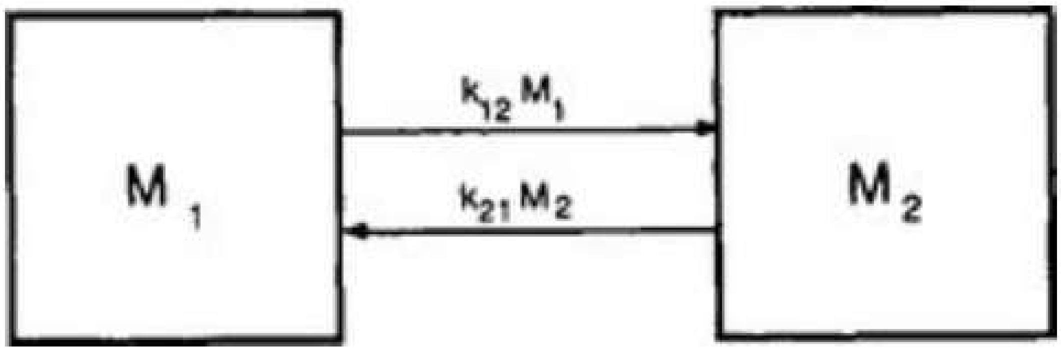


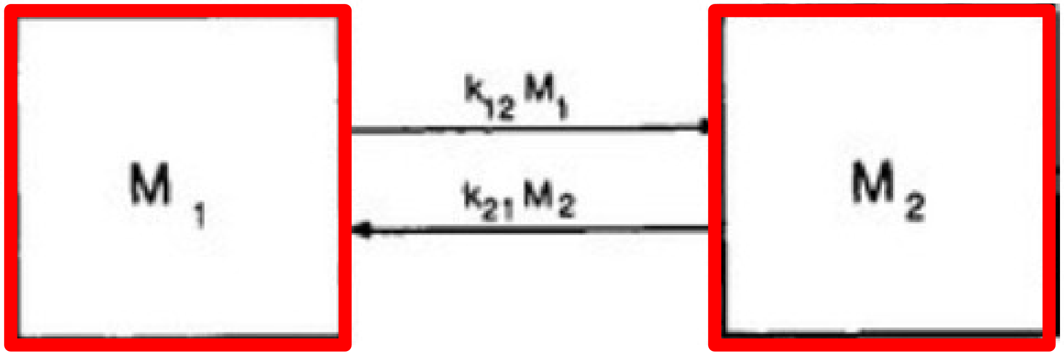
FIG. 4-6 A coupled two-reservoir system with fluxes proportional to the content of the emitting reservoirs.

- The response time of the system is defined as $\tau_{cycle} = \frac{1}{|E_1|}$

  where E_1 is the smallest nonzero eignevalue of the k_{ij} matrix

Linear coupled reservoir systems

$$\frac{dM_1}{dt} = -k_{12}M_1 + k_{21}M_2 \quad \frac{dM_2}{dt} = k_{12}M_1 - k_{21}M_2$$



The behavior of a coupled two-reservoir system with no external forcing is governed by two differential equations

The constraint of mass conservation in a closed system $M_t = M_1 + M_2$

and the initial conditions $M_1(t=0) = M_{10}$ & $M_2(t=0) = M_t - M_{10}$ apply.

The same thing written in matrix form looks like this: $\frac{dM}{dt} = [k_{ij}]\vec{M}$

and the k_{ij} matrix looks like this: $\begin{bmatrix} -k_{12} & k_{21} \\ k_{12} & -k_{21} \end{bmatrix}$

With respect to the k notation it is important to note that the first number in the subscript denotes the reservoir with whose mass the k is being multiplied and the second number in the subscript the reservoir that receives the mass. Both are each others source and sink.

Long version

$$\begin{bmatrix} -k_{12} & k_{21} \\ k_{12} & -k_{21} \end{bmatrix}$$

The eigenvector is a vector that has its direction unchanged (or reversed) by a given linear transformation. The corresponding eigenvalue is the multiplying factor λ used to scale it.

Step by step instruction of how to calculate it:

Step 1: Set up the equation

$$\begin{bmatrix} -k_{12}-\lambda & k_{21} \\ k_{12} & -k_{21}-\lambda \end{bmatrix} = (-k_{12}-\lambda)(-k_{21}-\lambda) = k_{12}k_{21} = 0$$

The solution for the eigenvalue is $\lambda_1=0$ and $\lambda_2=-(k_{12}+k_{21})$

Step 2: Solve the equation $\dfrac{dM}{dt} = [k_{ij}]\vec{M}$ **by integration**

The general solution is: $M_t = \psi_1 e^{\lambda_1 t} + \psi_2 e^{\lambda_2 t}$ **where ψ_1 and ψ_2 are the eigenvectors of matrix k. Since $e^0 = 1$ this becomes** $M_t = \psi_1 + \psi_2 e^{-(k_{12}+k_{21})t}$

Long version

$$M_t = \psi_1 + \psi_2 e^{-(k_{12}+k_{21})t}$$

written in component form and in terms of the initial conditions looks like:

$$M_1(t) = \frac{k_{21}}{k_{12}+k_{21}} M_t + \left(M_{10} - \frac{k_{21} M_t}{k_{12}+k_{21}}\right) e^{-(k_{12}+k_{21})t}$$

$$M_2(t) = \frac{k_{12}}{k_{12}+k_{21}} M_t + \left(M_T - M_{10} - \frac{k_{12} M_t}{k_{12}+k_{21}}\right) e^{-(k_{12}+k_{21})t}$$

Executive summary

The mass gets distributed between the two reservoirs in proportionality to the two sink coefficients in reverse proportion to the turnover times independent of how it was initially distributed and τ_{cycle} is the time it takes to do ao

$$\tau_{cycle} = \frac{1}{k_{12}+k_{21}}$$

Case study – open reservoir in steady state

- Material in introduced at a constant rate Q into reservoir 1. Some of this material is removed (S_1) and the rest (T) is transferred to reservoir 2 from which it is removed a the rate S_2

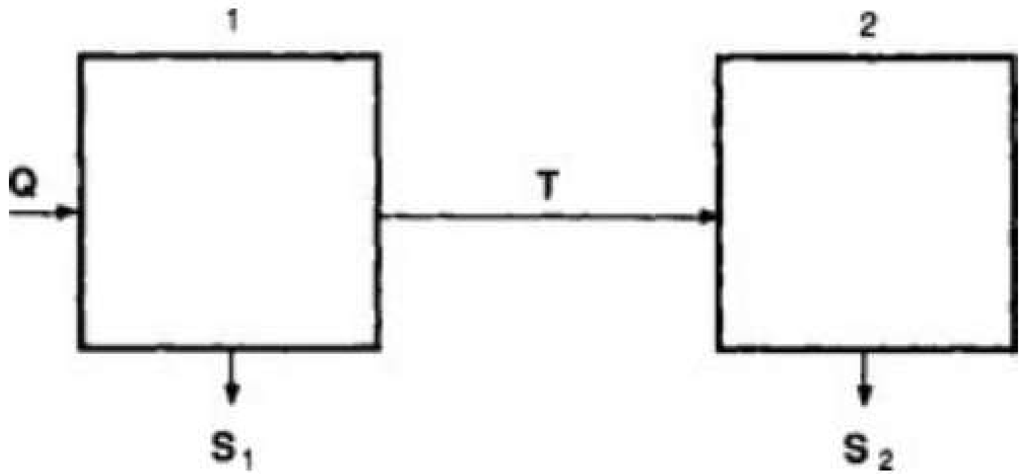


FIG. 4-9 An open two-reservoir system.

- The turnover time of the two individual reservoirs can be calculated to be

$$\tau_{01} = \frac{M_1}{T + S_1} = \frac{M_1}{Q} \qquad \tau_{02} = \frac{M_2}{S_2} = \frac{M_2}{T}$$

- The turnover time of the combined reservoir (sum of the two) is

$$\tau_0 = \frac{M_1 + M_2}{S_1 + S_2} = \frac{M_1 + M_2}{Q} = \frac{M_1}{Q} + \frac{M_2}{S_2} \cdot \frac{T}{Q} = \tau_{01} + \alpha\tau_{02}$$

Coupled reservoir systems

- **For systems with more than two reservoirs there is no unique mathematical solution but only one that depends on the initial conditions. In the case below specifically on how the mass was initially distributed between the top two reservoirs**

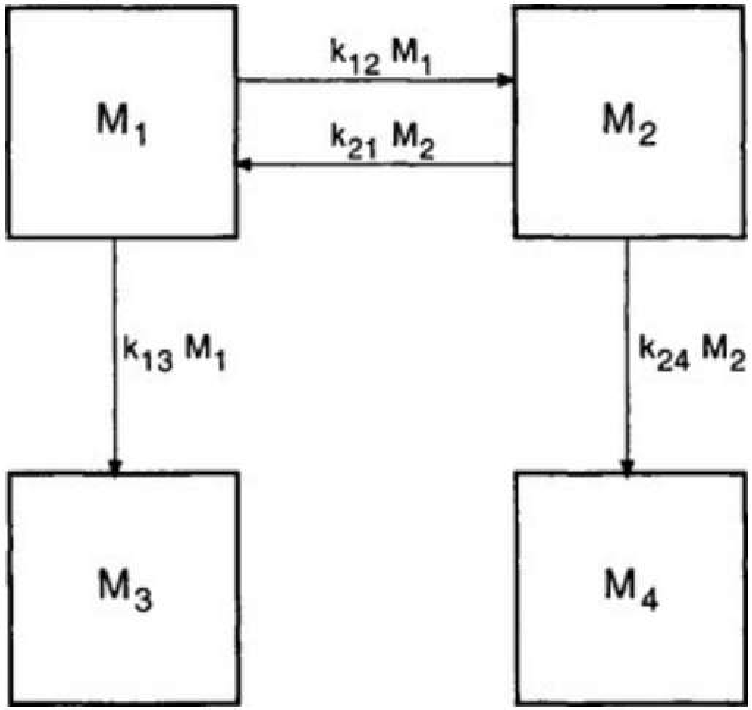


FIG. 4-7 Example of a coupled reservoir system where the steady-state distribution of mass is not uniquely determined by the parameters describing the fluxes within the system but also by the initial conditions (see text).

Case study

- **A model of the phosphorous cycle perturbed by exponentially increasing mining input growing by 7% p.a.**

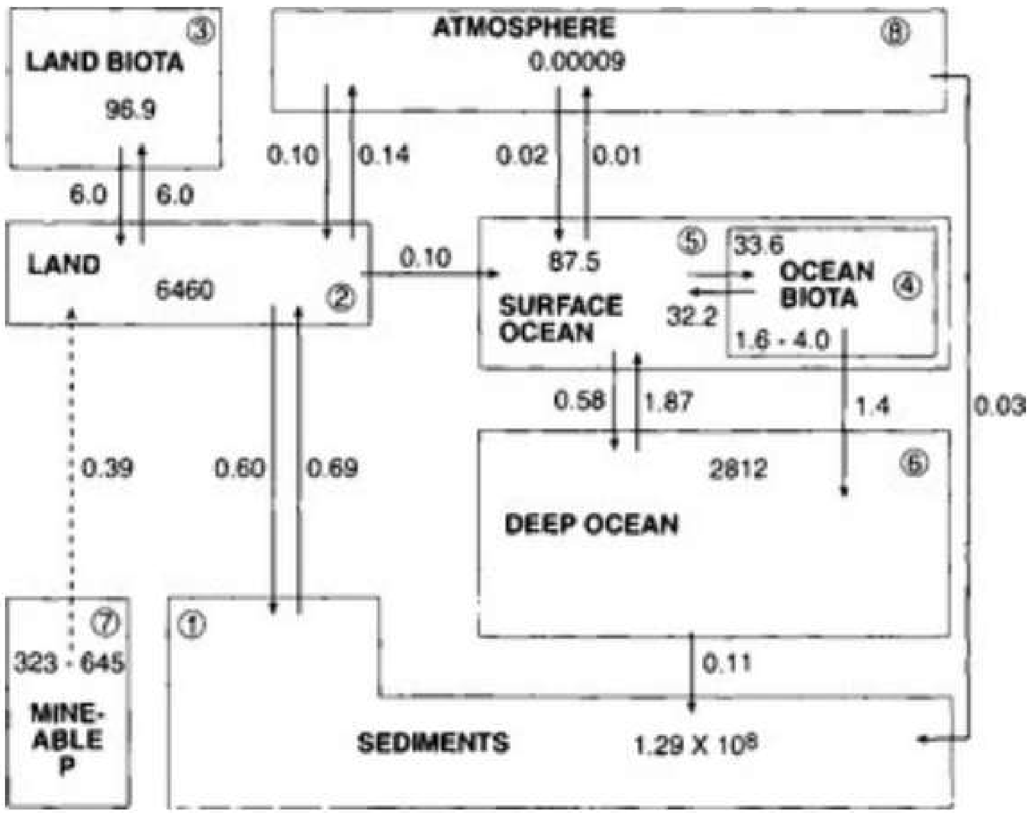


FIG. 4-8 The global phosphorus cycle. Values shown are in Tmol and Tmol/yr. (Adapted from Lerman *et al.* (1975) and modified to include atmospheric transfers. The mass of P in each reservoir and rates of exchange are taken from Jahnke (1992), MacKenzie *et al.* (1993) and Follmi (1996) .)

Case study

- With the adjustment process shown for the initial 70 years

Table 4-1

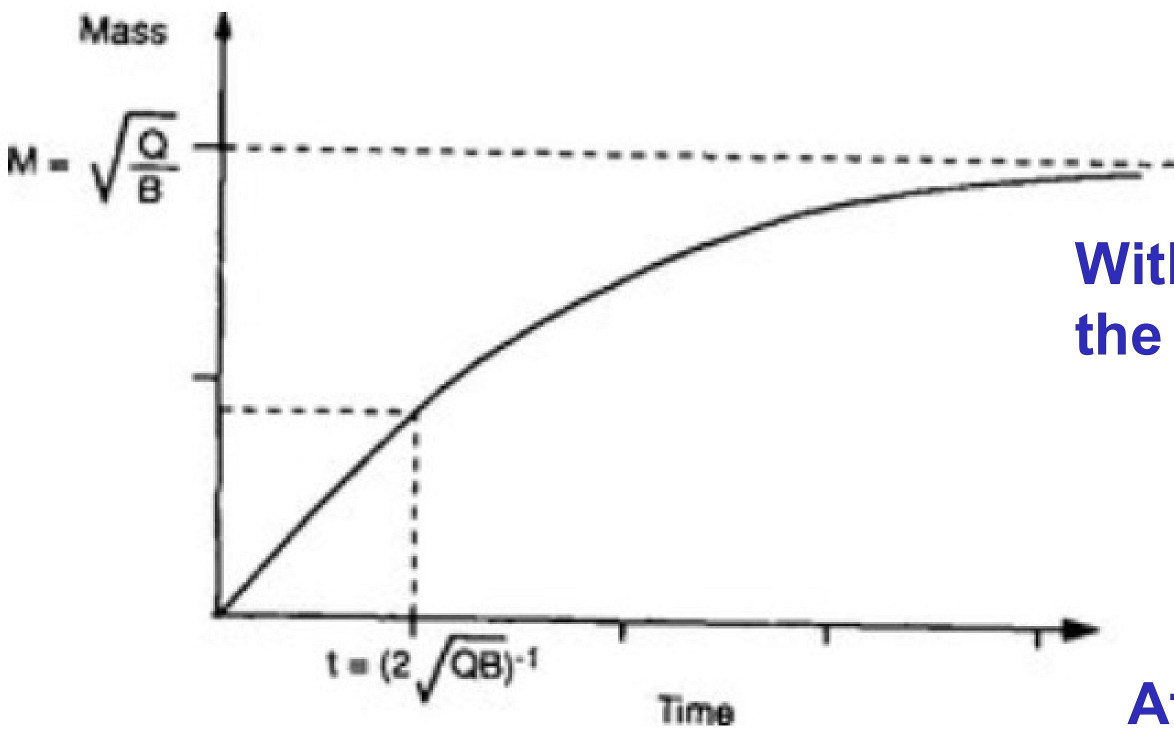
Response of phosphorus cycle to mining output. Phosphorus amounts are given in Tg P (1 Tg = 10^{12} g). Initial contents and fluxes as in Fig. 4-7 (system at steady state). In addition, a perturbation is introduced by the flux from reservoir 7 (mine-able phosphorus) to reservoir 2 (land phosphorus), which is given by 12 exp(0.07*t*) in units of Tg P/yr

Time (years)	Land	Land biota	Organic biota	Surface ocean	Deep ocean
0	200 000	3000	138	2710	87 100
10	200 173	3000	138	2710	87 100
20	200 522	3001	138	2710	87 100
30	201 224	3003	138	2710	87 100
40	202 636	3008	138	2710	87 100
50	205 481	3018	138	2710	87 100
60	211 208	3018	138	2711	87 100
70	222 741	3078	138	2712	87 101

- The only noticeable change is for the land & land biota as λ_{cycle} is 53,000 years, which is shorter than the turnover time of sediments $2x10^8$ years but longer than the turnover time of all other reservoirs.

Nonlinear systems

- A single reservoir with a constant source and a sink proportional to the square of the reservoir content.



The rate of change is:

$$\frac{dM}{dt} = Q - BM^2$$

With the initial conditions M(0)=0 the solution is

$$M_t = \sqrt{\frac{Q}{B} \cdot \frac{1 - e^{(-2\sqrt{QBt})}}{1 = e^{(-2\sqrt{QBt})}}}$$

After the time $\dfrac{1}{2\sqrt{QB}}$

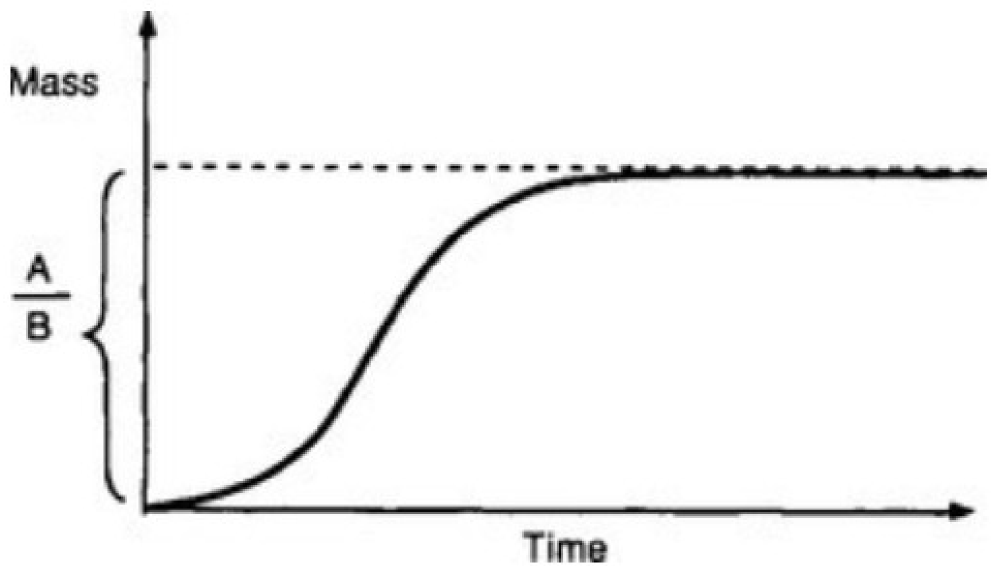
M reaches a steady state equal to $\sqrt{\dfrac{Q}{B}}$

But the adjustment is not uniform and slows down as time progresses

FIG. 4-10 The shape of the function given in Equation (28).

Nonlinear systems = logistic growth

- **You may be familiar with the logistic growth of biological systems**



The equation is

$$\frac{dM}{dt} = AM - BM^2$$

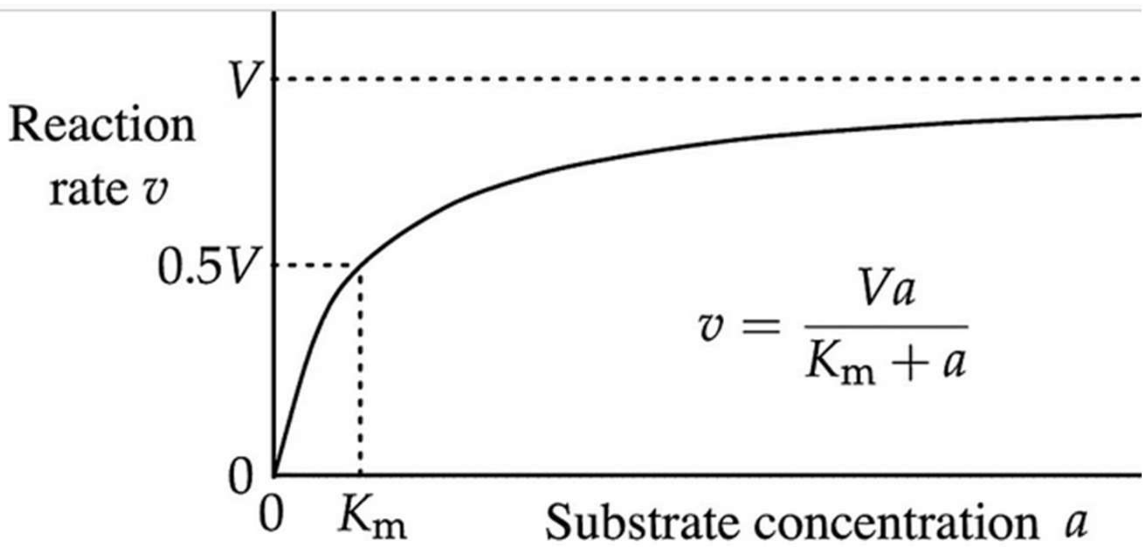
The term AM refers to a source term that is proportional to the mass already there and undergoes unlimited exponential growth with unlimited supply of space and nutrients and the –BM^2 term represents a negative feedback effect in the form of crowdedness. The mass levels off at A/B

FIG. 4-11 Shape of "logistical growth." The rate of change increases slowly initially. The rate of growth reaches a maximum and eventually drops to zero as the mass levels off, approaching the value *A/B.*

Nonlinear systems = Michaelis Menten equation

- You may also be familiar with the Michaelis Menten equation for enzyme catalyzed reaction where the sink flux has a less than proportional dependence on the mass

$$S = \frac{CM}{(M + D)}$$



C is the rate coefficient and D the Michaelis constant which determines the deviation from proportionality.

Nonlinear systems in the carbon cycle

- Both the exchange of CO_2 from the ocean to the atmosphere and from the atmosphere to the land are non-linear.

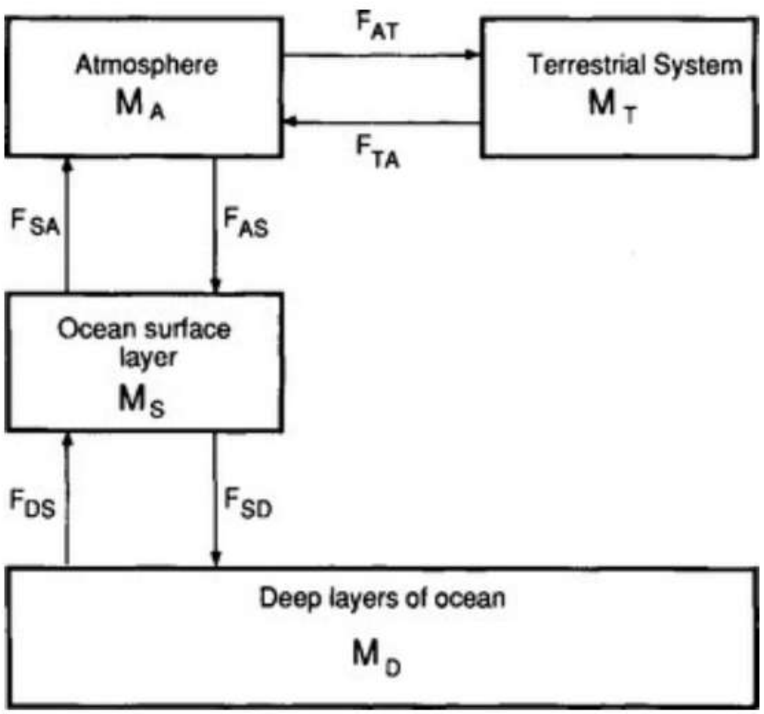


FIG. 4-12 Simplified model of the biogeochemical carbon cycle. (Adapted from Rodhe and Björkström (1979) with the permission of the Swedish Geophysical Society.)

In the case of the ocean M_s represents the sum of all carbon in the form of CO_2(aq), H_2CO_3, HCO_3^- and CO_2^{2-}. The flux from ocean to the atmosphere is dependent on the concentration of the dissolved CO_2(aq) which is related to the total carbon content in the surface layer by

$$F_{SA} = k_{SA} M_S^{\alpha_{SA}}$$

The exponent α_{SA} also known as buffer or Revelle factor is about 9

The buffer factor results from the equilibrium between CO_2(g) and the more prevalent forms of dissolved carbon. End result is a small net increase in M_s despite a massive increase in M_A

Nonlinear systems in the carbon cycle

- Both the exchange of CO_2 from the ocean to the atmosphere and from the atmosphere to the land are non-linear.

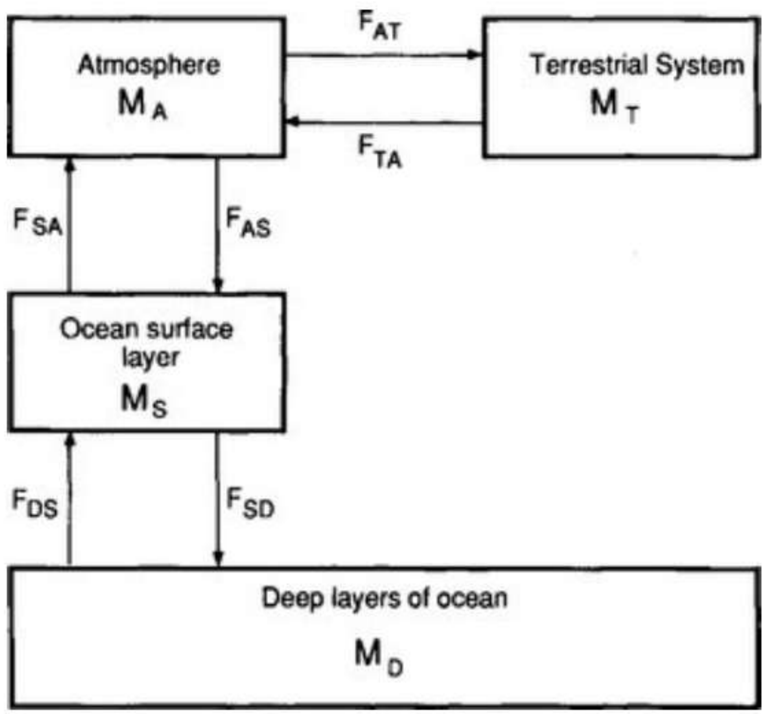


FIG. 4-12 Simplified model of the biogeochemical carbon cycle. (Adapted from Rodhe and Björkström (1979) with the permission of the Swedish Geophysical Society.)

The flux of CO_2 to the terrestrial system is also nonlinear and described by

$$F_{AT} = k_{AT} M_A^{\alpha_{AT}}$$

The exponent α_{AT} in this case is very small because growth of terrestrial biomass is usually limited by nutrients or water availability not by CO_2

Nonlinear systems in the carbon cycle

- Both the exchange of CO_2 from the ocean to the atmosphere and from the atmosphere to the land are non-linear. Hence the greatest accumulation is in the atmosphere only.

Table 4-2

Steady-state carbon contents (unit: Pg = 10^{15} g) for the four-reservoir model of Fig. 4-11: (a) during the unperturbed (pre-industrial) situation; (b) after the introduction of 1000 Pg carbon; and (c) after the introduction of 6000 Pg carbon

		After 1000 Pg		After 6000 Pg	
	Pre-industrial content (Pg)	Content (Pg)	% increase	Content (Pg)	% increase
Atmosphere	700	840	20	1880	170
Terrestrial system	3000	3110	4	3655	22
Ocean surface layer	1000	1020	2	1115	12
Deep ocean	35 000	35 730	2	39 050	12

Fluxes governed by the receiving reservoir

- There are some important situations in which a flux between two reservoirs is not only determined by the mass of the emitting reservoir but also by the mass of the receiving reservoir.
- Think of the uptake of CO_2 by the biosphere. The amount of the available biomass M_{TB} also matters for the flux.

$$F_{AT} = \frac{k_{ATB} M_A M_{TB}}{M_A + D}$$

- The equation above in which D as a Michaelis constant is a better representation of the flux than the one with simplifies it as α_{AT}

$$F_{AT} = \frac{k_{AT} M_A \left[\frac{M_{TB} \left(M_{TB_{max}} - M_{TB} \right)}{M_{TB_{max}}} \right]}{M_A + D}$$

- However, even more accurate is a version in which M_{TB} growth is logistic and limited by water, land and nutrient availability

Different forms of models used in studying biogeochemical cycles

- **Forward modeling:** Models are used to estimate the concentration or total mass in the various reservoirs based on information about sources and sink processes.
- **Model testing:** If direct measurements of the concentration of the substance in various reservoirs are available, then they can be compared to model predictions. If there are significant differences between observations and model simulations then improvements in the model are necessary.
- **Inverse modeling:** represents a situation when a model is used in a systematic fashion to estimate the magnitude of sources or sinks from observed concentrations (mass). In the simplest case of a single reservoir in steady state the formal solution follows directly from $Q=kM$. When concentrations vary in space and time or several reservoirs are involved, inverse models are more complex and statistical methods are often used to approximate a solution computationally.

Different forms of models used in studying biogeochemical cycles

- **Transport models:** Study the distribution within fluid reservoirs.
- **1D transport models:** look only at the distribution within the vertical column above a point. This is useful when the distribution in the horizontal direction is homogeneous and the main transport is in the vertical direction e.g. via detritus in the ocean.
- **2D transport models:** represent the height and the latitude but not the longitudinal transport. These can be used to study atmospheric circulation cells redistributing heat moisture and energy like the Hadley cell.
- **3D transport models:** represent transport in altitude, latitude and longitude. Weather forecast models and chemical transport models are an example. Their drawback is that they are computationally very complex. In such models all processes occurring at a lower spatial scale than the grid size which is often 100 km x 100 km must be parameterized if they are important for the net transport.

Different forms of models used in studying biogeochemical cycles

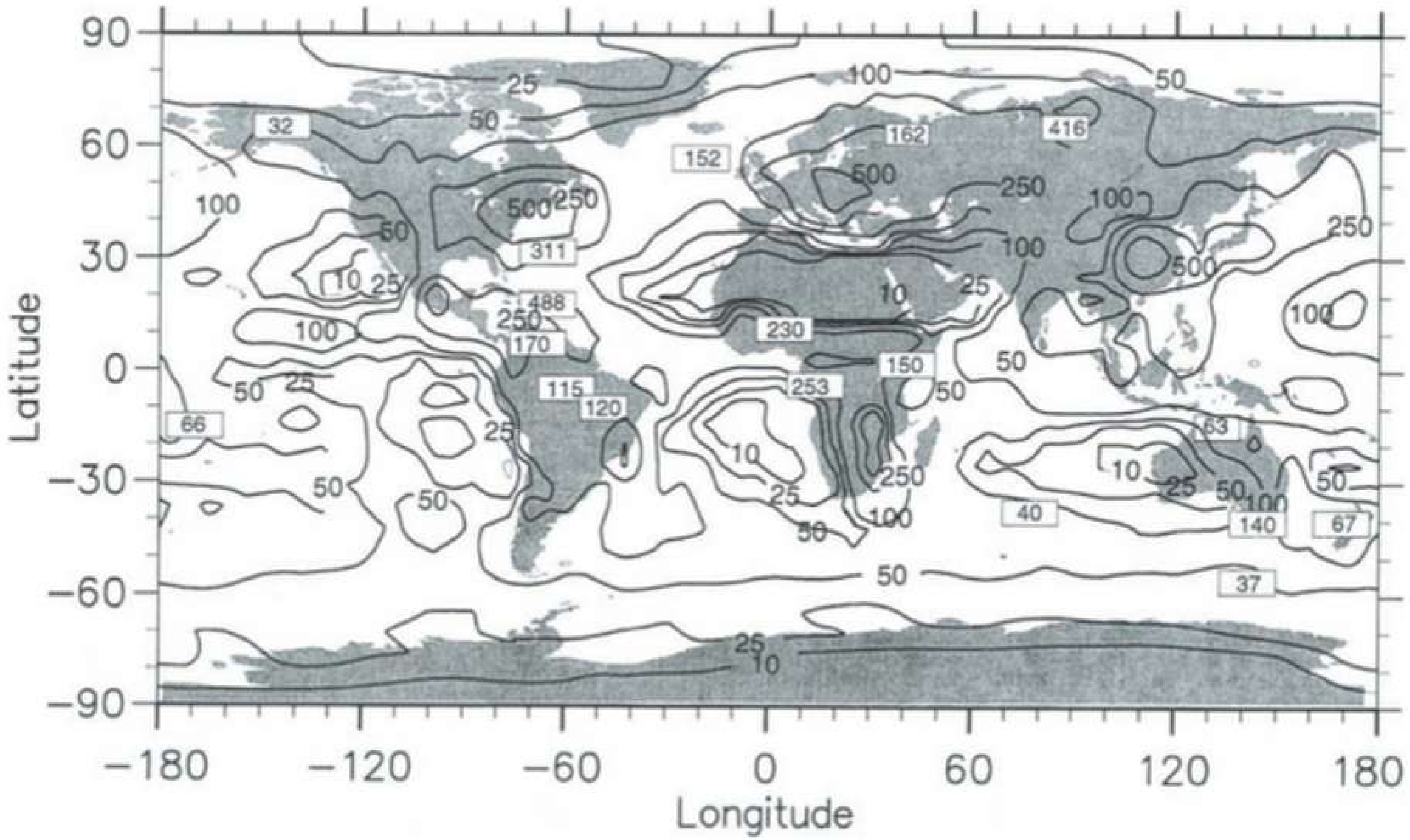


FIG. 4-13 Calculated and observed annual wet deposition of sulfur in mg S/m^2 per year. (Reprinted from "Atmospheric Environment," Volume 30, Feichter, J., Kjellström, E., Rodhe, H., Dentener, F., Lelieveld, J., and Roelofs, G.-J., Simulation of the tropospheric sulfur cycle in a global climate model, pp. 1693–1707, Copyright © 1996, with permission from Elsevier Science.)

Global Climate models

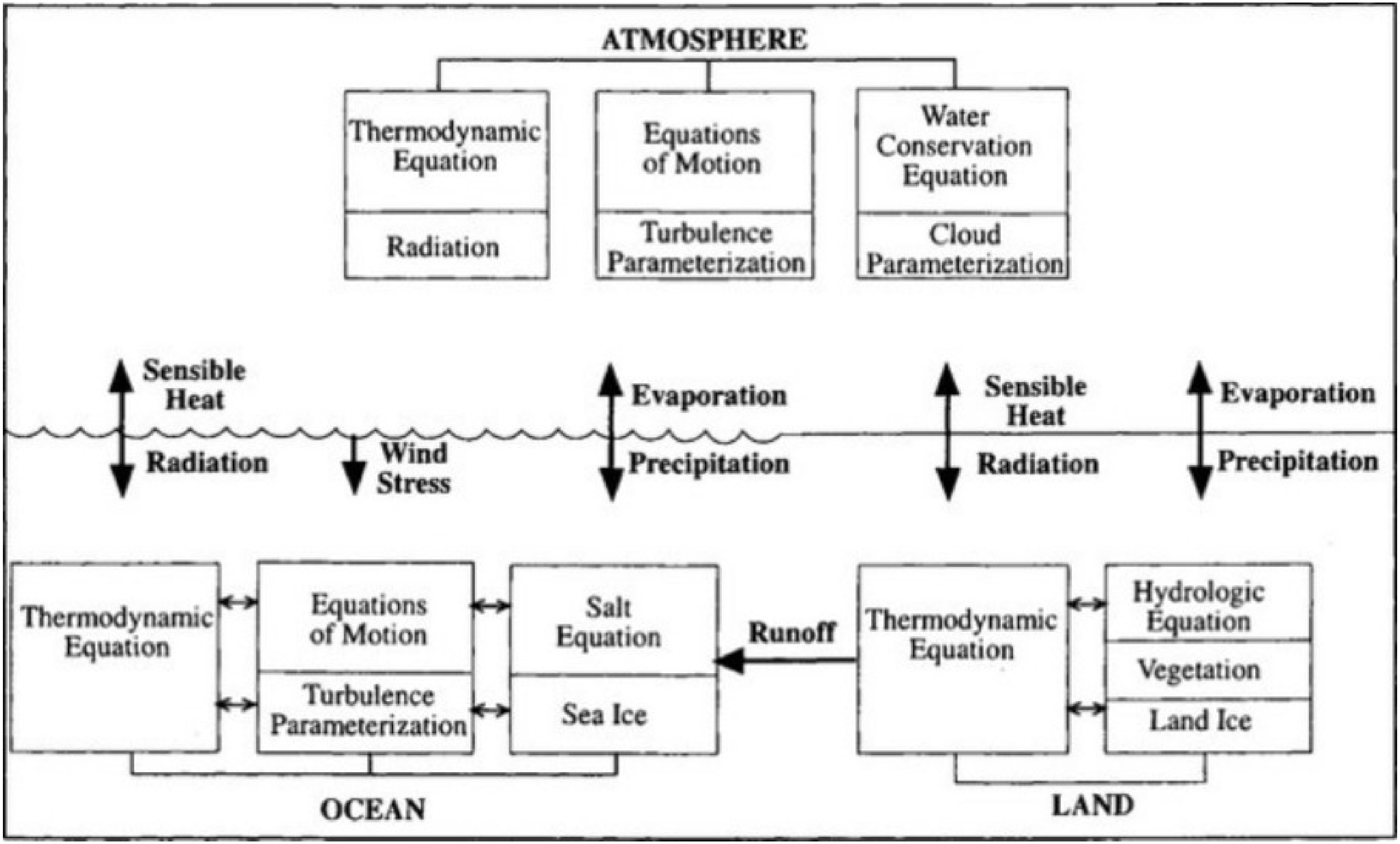
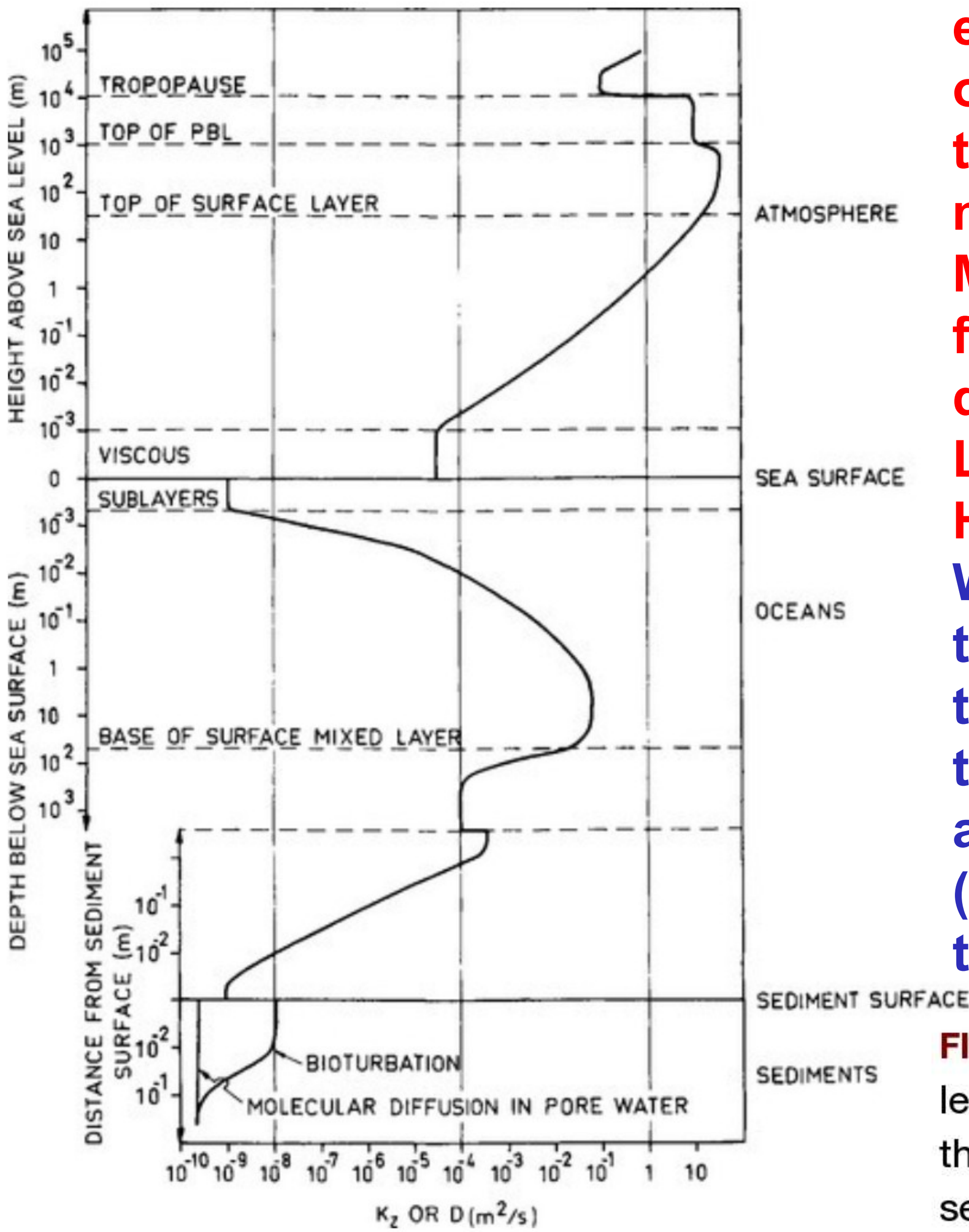


FIG. 4-14 Schematic diagram showing the components of a global climate model (GCM). (Reprinted from Hartmann (1994), with permission from Academic Press.)

Some important notes on transport



While transport via diffusion processes is easier to describe mathematically it is only relevant to the net flux in the tropopause and at interfaces where fluid motion is almost absent.

Most transport occurs through large scale fluid motion (winds, currents) not diffusion and via turbulent fluxes

Lateral transport is called advection.

Horizontal transport is called convection.

We are skipping the mathematical treatment of convection and advection. In this course we are happy to describe things at the scale of reservoirs (boxes) and transport between those boxes (fluxes). We don’t trouble ourselves with the motion within each box.

FIG. 4-15 Orders of magnitude of the average vertical molecular or turbulent diffusivity (whichever is largest) through the atmosphere, oceans, and uppermost layer of ocean sediments.

Timescales of mixing

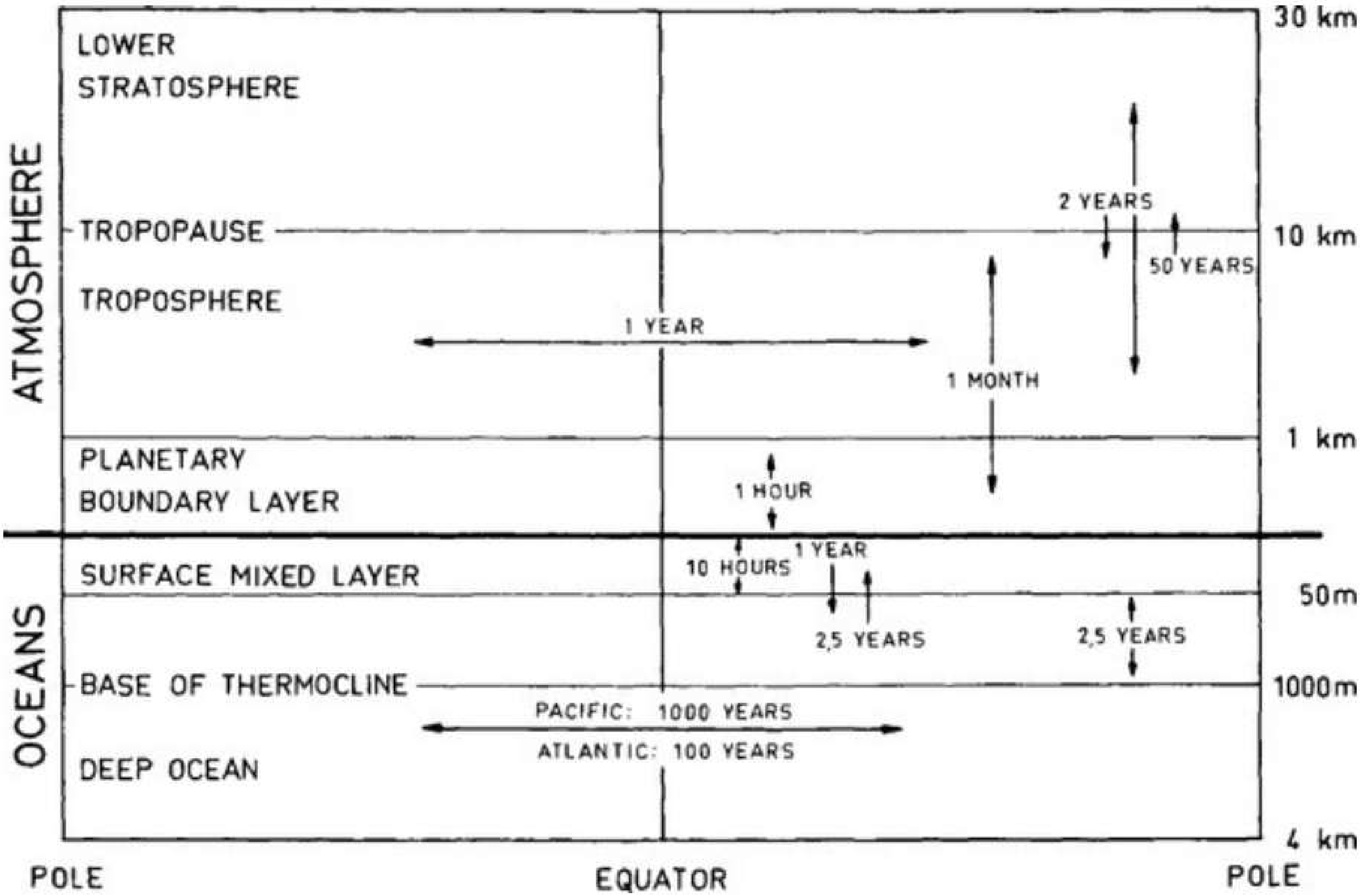


FIG. 4-17 Rough estimates of characteristic time for exchange of air and water respectively, between dfferent parts of the atmosphere and oceans.